

Lesson 06: Homogeneous & Bernoulli Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 6

Video: Homogeneous Differential Equations and Bernoulli Differential Equations **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that when standard methods fail, change the variables! In mathematics, substitution is the art of transforming an impossible problem into an easy one you already know how to solve. Homogeneous equations possess scale symmetry, allowing the radial substitution $y = vx$ to collapse them into separable equations. Bernoulli equations possess a nonlinear power term y^n , which can be neutralized by the substitution $u = y^{1-n}$ to convert them directly into linear first-order equations.

2. Core Theoretical Framework & Rigorous Formulations

Transformation Engines

1. Homogeneous Equations of Degree n :

If $M(tx, ty) = t^n M(x, y)$ and $N(tx, ty) = t^n N(x, y)$, substitute:

$$y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$$

This transforms the ODE into a separable equation in v and x .

2. Bernoulli Equations:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad (n \neq 0, 1)$$

Divide by y^n : $y^{-n}y' + P(x)y^{1-n} = Q(x)$. Substitute $u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$.

This yields the linear first-order equation in $u(x)$:

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$$

3. Fully Solved Whiteboard Problems

Concordia Midterm Problem: Solving a Bernoulli Equation

Problem Statement: Solve the nonlinear differential equation $\frac{dy}{dx} - \frac{1}{x}y = xy^2$ for $x > 0$.

Step 1: Identify Bernoulli Form and Exponent

$y' - \frac{1}{x}y = xy^2$. Here $n = 2$, $P(x) = -\frac{1}{x}$, and $Q(x) = x$.

Step 2: Linearizing Substitution

Divide by y^2 : $y^{-2}y' - \frac{1}{x}y^{-1} = x$. Let $u = y^{1-2} = y^{-1} \implies \frac{du}{dx} = -y^{-2} \frac{dy}{dx}$.

Step 3: Construct Linear Equation in u

Multiply by -1 : $-y^{-2}y' + \frac{1}{x}y^{-1} = -x \implies \frac{du}{dx} + \frac{1}{x}u = -x$.

Step 4: Solve via Integrating Factor

$\mu(x) = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$. Multiply across: $\frac{d}{dx}[xu] = -x^2 \implies xu = -\frac{x^3}{3} + C$.

Step 5: Back-Substitute $u = y^{-1}$

$u(x) = -\frac{x^2}{3} + \frac{C}{x} = \frac{3C - x^3}{3x} \implies y(x) = \frac{3x}{3C - x^3}$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

When computing $du/dx = (1-n)y^{-n}dy/dx$, students frequently forget the chain rule factor $(1-n)$, which introduces a disastrous sign error in the transformed ODE.

4. Key Takeaways & Exam Strategy Formula Card

 Core Takeaway & Exam Rule

Bernoulli equations are linearized by $u = y^{1-n}$. Homogeneous equations are separated by $y = vx$.

- Video reference: [Homogeneous Differential Equations and Bernoulli Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.